

EDUSAMBAM ASSESSMENT SERIES

Practice Test: Bearings, 3D Trigonometry & Graphing Functions

PRINTABLE PRACTICE WORKSHEET

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Subject	Mathematics (Maths)	Topic	Practice Test: Bearings, 3D Trigonometry & Graphing Functions	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20

Total	45	—	100
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Section A — Multiple Choice

30 marks

1. A bearing is measured:

Basic

- A. Anticlockwise from South
- C. Clockwise from East

- B. Clockwise from North
- D. In any direction, as long as it is consistent

Answer: ☐ A ☐ B ☐ C ☐ D

2. Bearings are always written using:

Basic

- A. Two figures
- C. Fractions of a full turn

- B. Compass letters only (N, S, E, W)
- D. Exactly three figures

Answer: ☐ A ☐ B ☐ C ☐ D

3. Due East as a bearing is:

Basic

- A. 000°
- C. 090°

- B. 180°
- D. 270°

Answer: ☐ A ☐ B ☐ C ☐ D

4. The bearing of B from A is 065°. The back bearing (A from B) is:

Intermediate

- A. 245°
- C. 115°

- B. 065°
- D. 295°

Answer: ☐ A ☐ B ☐ C ☐ D

5. A cuboid has length 3 cm, width 4 cm, and height 12 cm. Its base diagonal is:

Intermediate

- A. 7 cm
- C. 5 cm

- B. 12 cm
- D. 25 cm

Answer: ☐ A ☐ B ☐ C ☐ D

6. Using the cuboid above (base diagonal 5 cm, height 12 cm), the space diagonal is:

Advanced

- A. 13 cm
- C. 7 cm

- B. 17 cm
- D. 60 cm

Answer: ☐ A ☐ B ☐ C ☐ D

7. The correct method to find a cuboid's space diagonal is:

Intermediate

- A. Add length, width and height
- C. Multiply length by width by height

- B. Use the sine rule only
- D. Find the base diagonal first, then combine it with the height using Pythagoras again

Answer: ☐ A ☐ B ☐ C ☐ D

8. Plotted points on a quadratic graph should be joined with:

Basic

- A. Straight line segments
- C. Dots only, unjoined

- B. A single smooth curve
- D. A dashed line

Answer: ☐ A ☐ B ☐ C ☐ D

9. The shape of every quadratic graph is called a:

Basic

- A. Parabola
- C. S-curve

- B. Hyperbola
- D. Ellipse

Answer: ☐ A ☐ B ☐ C ☐ D

10. $y = (x - 4)^2 + 7$. The turning point is:

Intermediate

- A. (-4, 7)
- C. (4, 7)

- B. (4, -7)
- D. (-4, -7)

Answer: ☐ A ☐ B ☐ C ☐ D

11. $y = (x - 3)(x - 9)$. The roots are at:

Intermediate

A. $x = -3$ and $x = -9$

B. $x = 3$ and $x = 9$

C. $x = 3$ only

D. $x = 27$

Answer: ☐ A ☐ B ☐ C ☐ D

12. A cubic graph with a positive x^3 coefficient generally:

Intermediate

A. Falls from left to right

B. Is a closed loop

C. Is a straight line

D. Rises from bottom-left to top-right in an S-shape

Answer: ☐ A ☐ B ☐ C ☐ D

13. The solutions of $f(x) = 0$ can be found graphically by looking at:

Intermediate

A. The y-intercept

B. The turning point

C. The steepest point

D. Where the curve crosses the x-axis

Answer: ☐ A ☐ B ☐ C ☐ D

14. For a strict inequality such as $y > 2x + 1$, the boundary line should be drawn:

Intermediate

A. As a thick solid line

B. As a dashed line

C. As a curve

D. In a different colour only

Answer: ☐ A ☐ B ☐ C ☐ D

15. To decide which side of a boundary line to shade, you should:

Intermediate

A. Test a convenient point, such as the origin

B. Always shade above the line

C. Always shade below the line

D. Guess based on the shape of the line

Answer: ☐ A ☐ B ☐ C ☐ D

Section B — True / False

20 marks

16. Due South as a bearing is 180° .

Basic

☐ True ☐ False

17. The back bearing is found by adding 180° when the original bearing is already 180° or more.

Intermediate

☐ True ☐ False

18. Bearing problems are usually solved using the sine rule, cosine rule, or SOH-CAH-TOA, once a triangle is identified.

Intermediate

☐ True ☐ False

19. The space diagonal of a cuboid is the longest diagonal, from one corner through the interior to the opposite corner.

Basic

☐ True ☐ False

20. To find a cuboid's space diagonal, you should multiply length $\times$ width $\times$ height.

Intermediate

☐ True ☐ False

21. A cubic graph can have up to two turning points.

Intermediate

☐ True ☐ False

22. A solid boundary line means the inequality is strict ($<$ or $>$).

Intermediate

☐ True ☐ False

23. Completing the square directly reveals a quadratic's turning point.

Intermediate

☐ True ☐ False

24. The axis of symmetry of $y = (x - a)(x - b)$ lies exactly halfway between the roots.

Advanced

☐ True ☐ False

25. When several inequalities are combined on one graph, the answer region only needs to satisfy one of them.

Intermediate

☐ True ☐ False

Section C — Fill in the Blank

20 marks

26. A bearing is measured clockwise from _____.

Answer:

27. The bearing of D from C is 230° . The back bearing (C from D) is _____ degrees.

Answer:

28. To find the angle between a space diagonal and the base of a cuboid, you use the right-angled triangle formed by the base diagonal and the _____.

Answer:

29. A cuboid's space diagonal is found using Pythagoras' theorem applied _____ times.

Answer:

30. Writing a bearing as " 45° " instead of "_____" is a common mistake.

Answer:

31. Writing $y = (x - p)^2 + q$, the turning point is at $(p, \text{_____})$.

Answer:

32. A dashed boundary line on a graph means the inequality is _____.

Answer:

33. The point where a graph crosses the y-axis is called the _____.

Answer:

34. A parabola's line of symmetry is called the axis of _____.

Answer:

35. To solve $f(x) = 0$ graphically, you read the points where the curve crosses the _____-axis.

Answer:

Section D — Column Matching

10 marks

36–40. Match each term to its correct definition.

Column A	Ans.	Column B
Back bearing	_____	A. An S-shaped curve from a function with an x^3 term
Space diagonal	_____	B. The bearing of the return journey, found using $\pm 180^\circ$
Turning point	_____	C. Boundary line used for a strict inequality ($<$ or $>$)
Cubic graph	_____	D. The longest diagonal of a cuboid, running through the interior
Dashed line	_____	E. The minimum or maximum point of a curve

Section E — Short Answer

20 marks

41. Explain how to find the back bearing of a given bearing.

42. Describe the two-step method for finding the space diagonal of a cuboid.

43. Explain how completing the square helps you find the turning point of a quadratic graph.

44. Describe how to decide which side of an inequality's boundary line to shade.

45. Explain the difference between a solid and a dashed boundary line on a graphical inequality.

Candidate Signature: _____

Invigilator Signature: _____

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Marks Obtained: _____ / 100 Checked By: _____

Result / Grade: _____ Date Checked: _____

Practice Test: Bearings, 3D Trigonometry & Graphing Functions — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. B | |
| 2. D | 9. A |
| 3. C | 10. C |
| 4. A | 11. B |
| 5. C | 12. D |
| 6. A | 13. D |
| 7. D | 14. B |
| 8. B | 15. A |

Section B — True / False

- | | |
|-----------|-----------|
| 16. True | 21. True |
| 17. False | 22. False |
| 18. True | 23. True |
| 19. True | 24. True |
| 20. False | 25. False |

Section C — Fill in the Blank

- | | |
|------------|-----------------|
| 26. North | 31. q |
| 27. 50 | 32. strict |
| 28. height | 33. y-intercept |
| 29. twice | 34. symmetry |
| 30. 045 | 35. x |

Section D — Column Matching

- | | |
|------------------------|---------------------|
| 36. Back bearing — B | |
| 37. Space diagonal — D | 39. Cubic graph — A |
| 38. Turning point — E | 40. Dashed line — C |

Section E — Short Answer (Model Answer Keywords)

41. 180; add; subtract
42. base diagonal; Pythagoras; height
43. turning point; p,q; (x-p)
44. test point; origin; substitute
45. solid; included; dashed; strict